

NIC ——— **OTP**

IP — Location
MAC — Identification } Device

MAC

- Media access control
- It is a physical address
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- 12*4 = 48 bit length

NIC — MAC

48 bit

24 bit Vendor 24 bit User-IP

5 - Suzen

7C-F1-7E-69-49-83

Surfing

MAC

Vendor

IP

- Internet protocol
- It is a logical address
- It is in dotted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- 8*4 = 32 bit length
- IP = network + host

Configuration

IPv4 Structure

32 bit

SIP 4 2 1

0 0

255 255 255 255

Ending IP

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Broadcast R&D

Invalid IP

Types of Comm

- Unicast — one to one
- Multicast — one to group
- Broadcast — one to all

50.0.0.10 — A — N.H.H.H

128.10.226.176 — B — N.N.H.H

192.162.182.172 — C — N.N.N.H

50.0.0.10

N. H. H. H

Network = Street no

Host = Door no

192 · 192 · 192 · 192

N · N · N · H

How many n/w bits ?

24

How many host bit

8

How many Host octet ?

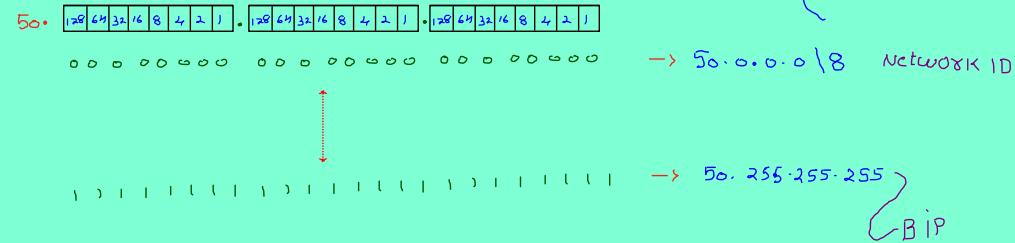
1

How many n/w octet ?

3

finding host IP in Per n/w

50.x.x.x



50.0.0.0/8

N/w ID

Permission

50.0.0.1 — fip
50.0.0.2
↓
50.0.0.255
50.0.1.0
50.0.1.1
50.0.1.2
↓
50.0.1.255
50.0.2.0
↓
50.0.255.255
50.1.0.0

50.255.255.254 — LIP
50.255.255.255 — BIP

formula

$$2^{24} = 16777216$$

$$-2$$

$$16777214$$

host bit

Insta → Firewall
50.0.0.0 — Block
N/w ID

HPN
50.1
50.2
50.3
Unicast
(50.255.255.255)

172.18.10.20
 N N H H
 172.18.X.X

$$2^{16} = 65,536$$

$$\underline{-2}$$

$$65,534$$

172.18.0.0

00000000 00000000

172.18.0.0 \ 16

172.18.0.1

172.18.255.254

172.18.255.255

11111111 11111111

192.192.192.192

192.192.192.0

00000000

192.192.192.0 \ 24

192.192.192.1

192.192.192.254

192.192.192.255

11111111

$$2^8 = 256$$

$$\underline{-2}$$

$$254$$

50.0.0.0

Subnetmask

It is use to mention n/w position

Condition

Put binary combination as 1 for n/w bits

50.0.0.10

00000001

11111111

172.16.0.10

255.0.0.0

255.255.0.0

IP & mac ✓

IP Parameter ✓

172.80.172.86

